

Frequency-Domain Refinement of Vision Transformers for Robust Medical Image Segmentation under Degradation

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Abstract

Medical image segmentation is crucial for precise diagnosis, treatment planning, and disease monitoring in clinical settings. While convolutional neural networks (CNNs) have achieved remarkable success, they struggle with modeling long-range dependencies. Vision Transformers (ViTs) address this limitation by leveraging self-attention mechanisms to capture global contextual information. However, ViTs often fall short in local feature description, which is crucial for precise segmentation. To address this issue, we reformulate self-attention in the frequency domain to enhance both local and global feature representation.

Our approach, the Enhanced Wave Vision Transformer (EW-ViT), incorporates wavelet decomposition within the self-attention block to adaptively refine feature representation in low and high-frequency components. We also introduce the Prompt-Guided High-Frequency Refiner (PGHFR) module to handle image degradation, which mainly affects high-frequency components. This module uses implicit prompts to encode degradation-specific information and adjust high-frequency representations accordingly. Additionally, we apply a contrastive learning strategy to maintain feature consistency and ensure robustness against noise, leading to state-of-the-art (SOTA) performance in medical image segmentation, especially under various conditions of degradation. Source code is available at [GitHub](#).

1. Introduction

Medical image segmentation plays a pivotal role in disease diagnosis and quantitative assessment throughout the clinical workflow. Convolutional neural networks (CNNs) have emerged as pioneering approaches in this domain. The seminal U-Net [23], with its series of convolutional and down-sampling layers designed to gather contextual infor-

mation through a symmetrical hierarchical architecture, has demonstrated remarkable segmentation capabilities. Despite their widespread use, CNNs struggle to effectively model long-range dependencies due to their reliance on stacks of convolutional blocks to increase receptive fields.

Emergence of Vision Transformers (ViTs): To address these limitations, the ViT [10] was introduced. ViTs leverage the self-attention mechanism, excelling at capturing global contextual information and modeling long-range dependencies, thereby shifting the field towards transformer-based approaches in various vision tasks [14]. However, ViTs often exhibit weaker local feature description and lack of inductive bias compared to CNNs, which is critical for semantic segmentation in medical images.

Addressing ViT Deficiencies: Several methods have been proposed to mitigate the deficiencies of ViTs in local feature capturing [19, 33] by applying convolutional blocks to capture local contextual details. TransUNet [8] is the pioneering model to combine transformer and U-Net for medical image segmentation, utilizing Transformer blocks in the encoder to acquire global information and a convolutional backbone to obtain local details. HiFormer [11] integrates a Swin Transformer module and a CNN-based encoder to generate two multi-scale feature representations, which are fused via a Double-Level Fusion module.

Hybrid Architectures and their Limitation: Other subsequent models, such as TransBTS [31], CoTr [34], and TMUnet [2], also employ a CNN-based network as the backbone while incorporating the Transformer module to handle long-range dependencies, classifying them as hybrid transformer architectures. Despite their innovative designs, hybrid architectures have notable limitations. The TransUNet model, for instance, is characterized by a high number of parameters and computational inefficiency. Models like HiFormer [11] and the contextual network effectively merge a CNN and a Transformer for medical image seg-

mentation but still heavily depend on a CNN backbone. Furthermore, even with novel feature fusion methods that blend CNN and Transformer multi-scale features, there is insufficient consideration of multi-scale information. This inadequacy reduces segmentation accuracy due to the poor integration of high-level and low-level features extracted from the Transformer and convolutional backbones.

Drawbacks of Transformers: Despite efforts to mitigate the local feature deficiency in ViT models, integrating convolutional modules remains essential. Delving into the underlying reasons, studies [30] indicate that self-attention functions as a low-pass filter when Transformer blocks are properly stacked. High-frequency components play a vital role in many applications [35], providing fine-grained details crucial for vision-based tasks and precise localization [4]. However, transformer models predominantly capture low-frequency representations, making it difficult to extract high-frequency components [30].

Our Contributions: To address this challenge, leveraging the frequency domain has emerged as a promising approach. Recent methods [4, 5, 38] have integrated frequency-based techniques to preserve high-frequency information. While these methods effectively maintain high-frequency details, they often struggle when input images contain degradation. Medical images frequently suffer from noise, blur, and haze, impacting the quality of high-frequency components and resulting in poor localization and weak boundary delineation. Therefore, the transformer model requires a mechanism to handle such degradation.

In this regard, we model the transformer block using wavelet decomposition. Integrating wavelet analysis to enhance high-frequency representations in a ViT can provide a multi-resolution decomposition of input data, allowing the identification and recalibration of high-frequency components. To ensure noise-tolerant representation, our approach introduces the Prompt-Guided High-Frequency Refiner (PGHFR) module on top of the high-frequency component to implicitly learn the degradation from the image itself and account for reducing such noise from the representational space. Specifically, our method uses prompts to encode high-frequency degradation-specific information, which is then used to dynamically adjust high-frequency information for effective representation. This allows our method to generalize to different degradation types and levels while still achieving state-of-the-art results in image segmentation tasks. To further ensure feature consistency and implicitly guide the PGHFR module, we create different feature projections by manipulating the high-frequency information and employing a contrastive loss function to learn feature matching regardless of noise effects. This approach ensures that the model representation remains robust against possible degradation in the high-frequency components. Our contributions are as follows: (1) Wavelet-Based

Frequency Domain Modeling: We model the transformer within the frequency domain using wavelets, which reduces the computational complexity of self-attention by resulting in lower spatial dimensions while capturing multi-scale representations simultaneously. (2) Introduction of the PGHFR Module: We present the PGHFR module, which refines high-frequency components and addresses noise effects in the representation. (3) Contrastive Learning for feature consistency: A contrastive learning method to ensure feature consistency and provide guidance for the PGHFR module. (4) Evaluating degradation resilience: An extensive ablation study to showcase the robustness of our method against different degradation type. Implementation code along with the configuration will be available upon acceptance.

2. Methodology

This work introduces an end-to-end approach that incorporates frequency analysis into a multi-scale representation within the ViT framework. The architecture of the proposed method is illustrated in Figure 1. The model adopts a U-shaped configuration consisting of an encoder and two heads, a contrastive head and a segmentation head. The encoder is structured with three layers, each featuring an Enhanced Wave ViT (EW-ViT) followed by patch embedding. Correspondingly, the decoder is composed of three layers with EW-ViT and patch expansion. The patch embedding layers are responsible for generating hierarchical feature representations by performing downsampling and increasing dimensionality, while the patch expansion layers handle upsampling [12]. In the decoder, our PGHFR module within EW-ViT (Figure 2) refines high-frequency components and counteracts degradation effects on feature representation. The contrastive head ensures feature consistency and furnishes direction for the PGHFR module.

2.1. Enhanced Wave ViT (EW-ViT)

Preserving high-frequency information is crucial for accurate medical image segmentation, as it captures fine details and edges essential for precise tasks. This is particularly important in medical imaging, where subtle differences can significantly impact diagnosis and treatment planning. To accomplish this, we use wavelet decomposition to enhance the transformer’s ability to retain high-frequency components (e.g., boundary information/edges) during segmentation. Wavelet decomposition breaks down images into high and low-frequency components, providing opportunity for multi-resolution analysis. This approach detects both global and local features, enhances edge detection and feature extraction, and reduces noise, resulting in more accurate segmentation of medical images.

Formally, consider a 2D feature map $X \in \mathbb{R}^{H \times W \times D}$. First, we apply a linear transformation to reduce the channel dimension, producing $\tilde{X} = XW_d$, where $W_d \in \mathbb{R}^{D \times \frac{D}{4}}$.

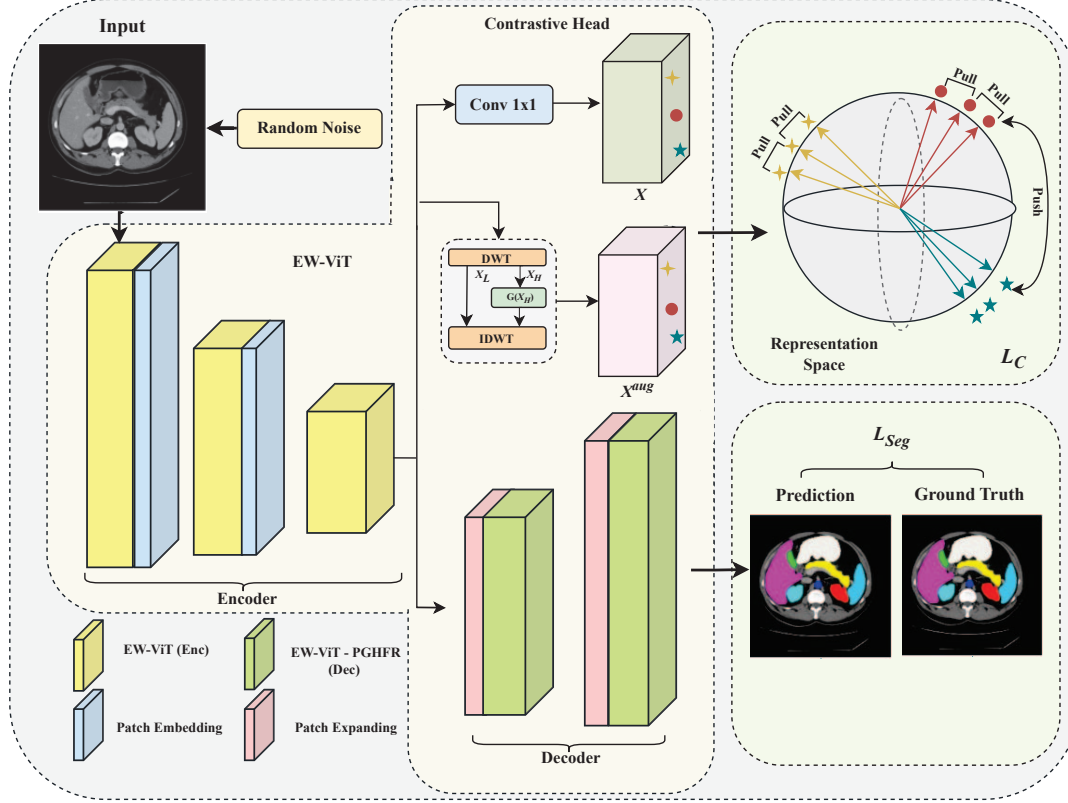


Figure 1. An illustration of our proposed EW-ViT. Initially, the input is processed through the *encoder* module. This process generates a parallel two-part structure. One part, managed by the *consistency head*, creates a representation space using a contrastive learning approach, where the $\mathcal{L}_C$ value is calculated. Concurrently, in the other part, the input image is processed through the *decoder* structure, equipped with *PGHFR*, resulting in a segmented image that is compared with the ground truth image using the $\mathcal{L}_{Seg}$ value.

Next, we use the Discrete Wavelet Transform (DWT) to down-sample $\tilde{X} \in \mathbb{R}^{H \times W \times \frac{D}{4}}$, decomposing it into four wavelet subbands. For simplicity, we use the classical Haar wavelet as described in [18]. Specifically, DWT applies the low-pass filter and the high-pass filter along the rows of $\tilde{X}$, resulting in two subbands: X_L and X_H . These filters are then applied along the columns of X_L and X_H , producing four wavelet subbands: $\hat{X} = [X_{LL}, X_{LH}, X_{HL}, X_{HH}] \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times \frac{D}{4}}$. The subband X_{LL} captures the low-frequency components, representing the basic structure of the object at a coarse-grained level. The subbands X_{LH} , X_{HL} , and X_{HH} capture the high-frequency components, retaining the fine-grained texture details. Thus, each wavelet subband is a down-sampled version of $\tilde{X}$, preserving all input details without any information loss.

However, image degradation poses significant challenges for frequency models by compromising high-frequency components, leading to poor localization and weak boundary delineation. To address these issues, we introduce PGHFR, which functions as a transformation $\hat{F}$. This transformation aims to reduce the effect of noise N on the high-frequency representation subbands $X_H =$

$[X_{LH}, X_{HL}, X_{HH}] \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times D'}$, where $[\cdot]$ stands for channel-wise concatenation and $D' = \frac{3D}{4}$. Specifically, $\hat{F}(X_H, N)$ dynamically adjusts X_H based on degradation-specific information (N) encoded in prompts, thereby improving localization, boundary delineation, and overall robustness of the model, resulting in X'_H .

A 3×3 convolution is applied to $[X_L, X'_H]$ to enforce spatial locality, resulting in the locally down-sampled feature map X^d . This feature map X^d is then linearly transformed into keys $K \in \mathbb{R}^{H/2 \times W/2 \times D}$ and values $V \in \mathbb{R}^{H/2 \times W/2 \times D}$. The EW-ViT's multi-head self-attention module, denoted as **Attention**, is performed over the queries ($Q = \text{Linear}(X) \in \mathbb{R}^{H \times W \times D}$) and their corresponding down-sampled keys and values for each head:

$$\text{head}_j = \text{Attention}(Q_j, K_j, V_j) = \text{Softmax} \left(\frac{Q_j K_j^T}{\sqrt{D_h}} \right) V_j, \quad (1)$$

where K_j , and V_j denote the keys and values for the j -th head, respectively. Moreover, we additionally apply inverse DWT (IDWT) over the channel-wise concatenation of

locally contextualized recalibrated high-frequency components X'_H , which are derived from the PGHFR, along with the low-frequency component X_L . Finally, we process all the contextualized information from each head and the recalibrated locally contextualized information, followed by a linear transformation to compose the outputs of our EW-ViT block as described below:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_0, \dots, \text{head}_{N_h}) \widetilde{W}^O, \quad (2)$$

where $\widetilde{W}^O$ is the transformation matrix.

2.2. Prompt-Guided High-Frequency Refiner (PGHFR)

Prompt-based techniques have gained significant traction in natural language processing (NLP) and computer vision due to their efficiency in fine-tuning large pre-trained models for specific tasks [17, 22, 25]. Traditionally, a "prompt" refers to an explicitly designed input, such as text or instructions, [13, 16, 36], that guides a model's response. However, recent advancements in prompt learning have expanded this concept to include learnable prompts [37], which adapt dynamically based on the input features. These learnable prompts are trainable parameters that interact with the model's internal representations, enabling efficient task-specific adaptations. In our proposed PGHFR modules within the EW-ViT framework (inspired by [20, 22, 28]), prompts are implemented as adaptive, learnable parameters. Rather than serving as static input queries, they interact directly with the input features to dynamically mitigate degradation in high-frequency components (as illustrated in Figure 2). This approach allows the model to refine feature representations in a context-aware manner, enhancing both localization precision and segmentation performance. Specifically, with N prompt components $P_c \in \mathbb{R}^{N \times \frac{H}{2} \times \frac{W}{2} \times \hat{D}'}$ and input features $X_H \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times D'}$, the function within the PGHFR block is defined as follows:

$$X'_H = \text{IPI}(\text{CPG}(P_c, X_H), X_H). \quad (3)$$

Our PGHFR module is composed of two main components: the Component Prompt Generation module (CPG) and the Implicit Prompt Interaction module (IPI), each of which will be described in detail in the following sections.

2.2.1 Component Prompt Generation (CPG)

Prompt components P_c interact with concatenated high-frequency subband features via learnable parameters, enhancing data processing and reducing noise effects. The CPG module dynamically predicts context-aware weights from input features to create input-conditioned prompts P

and facilitates shared knowledge among them. CPG processes input features X_H using Global Average Pooling (GAP) and Global Max Pooling (GMP) to capture spatial information. Their combined outputs are compressed through pixel-wise convolution (PWC), followed by a softmax operation to generate prompt weights $w \in \mathbb{R}^{D'}$, adjusting prompt components based on input relevance. Finally, a 3×3 convolutional layer refines features and enhances spatial relationships, improving task performance. Overall, the CPG process is summarized as follows:

$$\begin{aligned} \mathbf{w}_i &= \text{Softmax}(\text{PWC}(\text{GAP}(X_H) + \text{GMP}(X_H))). \\ P &= \text{Conv}_{3 \times 3} \left(\sum_{c=1}^{D'} \mathbf{w}_i P_c \right), \end{aligned} \quad (4)$$

2.2.2 Implicit Prompt Interaction (IPI)

The primary goal of the IPI module is to enable guided restoration by facilitating the interaction between the concatenated high-frequency input components X_H and the generated prompts P . This process enhances the representation of high-frequency components while reducing noise effects. In the IPI module, the generated prompts (P) are first concatenated channel-wise with the input components (X_H) and then processed through the **Attention** block. This recalibration aims to mitigate noise, ensuring robust segmentation and precise localization. The overall process of the IPI is described by:

$$X'_H = \text{Conv}_{3 \times 3}(\text{Attention}(X_H, P)). \quad (5)$$

2.3. Contrastive Approach

We implemented a deep contrastive supervision to refine model representation and enhance discriminative capabilities. In the bottleneck layer of our EW-ViT, we redesigned the architecture to implicitly identify degradation types and filter high-frequency channels. To strengthen the network against noise, we formulated a contrastive loss function. In this respect, we modeled the impact of random degradation noise on the high-frequency component as $G(X_H)$. Next, this was followed by IDWT to reconstruct an augmented input feature (X^{aug}). By resampling points from identical spatial locations in both the original and augmented inputs, we defined candidate pixels for contrastive learning. The class prototype in the contrastive space was computed as the mean vector of pixels belonging to the same class:

$$\mathbf{c}_k = \frac{1}{|S_k|} \sum_{i \in S_k} \mathbf{X}_i, \quad (6)$$

where $\mathbf{c}_k$ represents the prototype of class k and $\mathbf{X}_i$ is the feature representation of pixel i corresponding to class k .

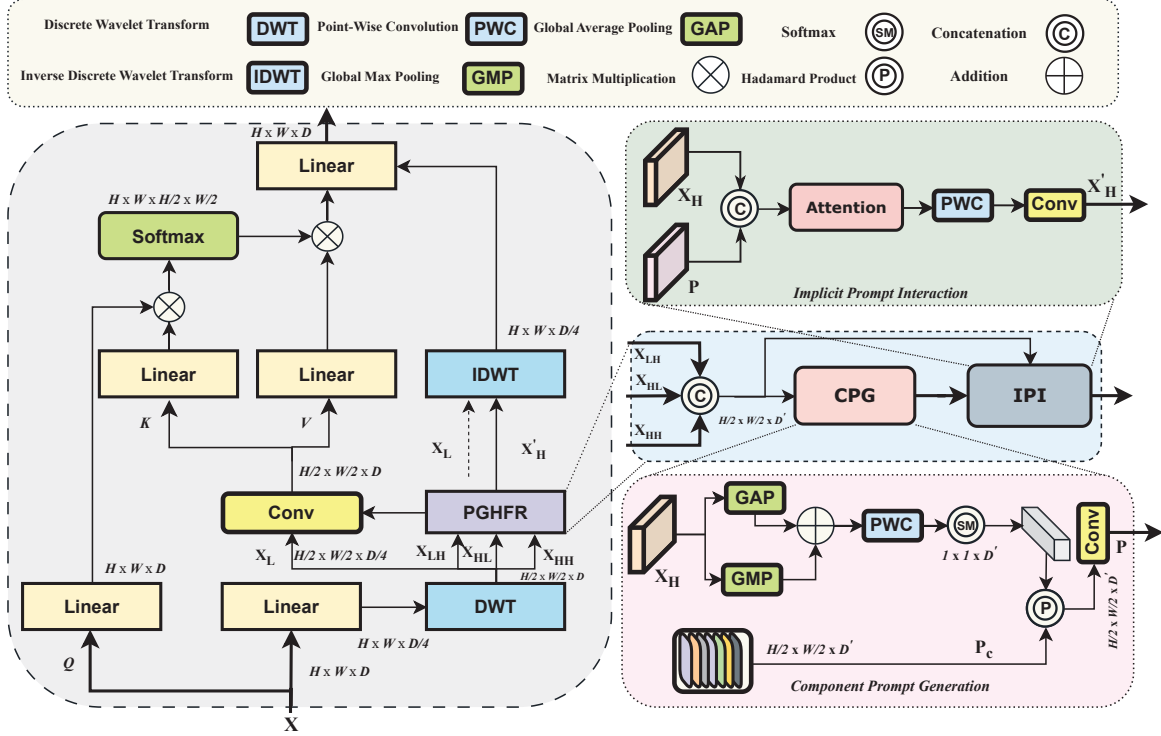


Figure 2. Overview of the **Enhanced Wave Attention Block**: $[X_{LL}, X_{HL}, X_{LH}, X_{HH}]$ represents the low-frequency, horizontal, vertical, and diagonal high-frequency components, respectively. The Prompt Block consists of two modules: CPG and IPI. The CPG module generates high-frequency-conditioned prompts (P) using concatenated features (X_H) and learnable parameters (P_c). The IPI module recalibrates the initial high-frequency components (X_H) using the generated prompts through attention.

S_k indicates the set of pixels that belong to class k . For the augmented candidates, we aligned them with the corresponding class prototype from X using a contrastive learning algorithm. The contrastive loss, designed to enhance the network’s predictions, was computed as follows:

$$\mathcal{L}_{c_k} = -\frac{1}{|S_k|} \sum_{(x_k^{aug}) \in S_k} \log \frac{\exp\left(\frac{\text{sim}(x_k^{aug}, c_k)}{\tau}\right)}{\sum_{j \neq k} \exp\left(\frac{\text{sim}(x_k^{aug}, c_j)}{\tau}\right)} \quad (7)$$

where $\text{sim}(\mathbf{X}_k^{\text{aug}}, \mathbf{c}_k)$ is the cosine similarity between a candidate pixel from the augmented image representation and the class prototype, and τ is a temperature parameter. Specifically, the overall contrastive loss $\mathcal{L}_c$ is the sum of the individual losses for each class k : $\mathcal{L}_c = \sum_k \mathcal{L}_{c_k}$. We also added a class prototype separation term, $\mathcal{L}_{sep} = \sum_{i \neq j} \frac{1}{\|\mathbf{c}_i - \mathbf{c}_j\|_2}$, to $\mathcal{L}_c$ to improve cluster separation.

3. Experiments

3.1. Datasets

Synapse Dataset: *Synapse* dataset [15] is utilized, which comprises 30 cases of abdominal CT scans. This dataset in-

cludes a total of 3,779 axial contrast-enhanced clinical CT images from the abdominal region. Each CT volume includes 85 to 198 slices of consistent size, 512×512 pixels in-plane, annotated for eight organs (i.e., multi-object segmentation). We adopted the data preparation from [8].

ISIC Dataset: In addition, we evaluate our method on dermatology images to segment skin lesion regions using the *ISIC2018* dataset [9], provided by the International Skin Imaging Collaboration (ISIC). We followed the evaluation criteria set forth in [1].

ACDC Dataset: Furthermore, we selected the *ACDC* dataset, which involves 100 cardiac MRI scans sourced from various patients. Each scan captures three specific organs: the right ventricle (RV), the left ventricle (LV), and the myocardium (Myo). Consistent with the methodologies outlined in [8], we utilized 70 cases (comprising 1,930 axial slices) for training, designated 10 for validation, and reserved 20 for testing.

3.2. Implementation Details

Our proposed method was implemented using the PyTorch library and experiments were executed on a single RTX 3090 GPU. For the *Synapse* dataset, we trained the model with a batch size of 12 and employed an SGD op-

Table 1. Evaluation results on the Synapse dataset: **blue** for the best and **red** for the second best. Metrics include parameters (in millions) and FLOPS (in billions). The overall DSC and HD95 values are reported, along with the DSC for various abdominal organs: spleen (Spl), right kidney (RKid), left kidney (LKid), gallbladder (Gal), liver (Liv), stomach (Sto), aorta (Aor), and pancreas (Pan).

Methods	Params (M)	FLOPs (G)	Spl.	RKid.	LKid.	Gal.	Liv.	Sto.	Aor.	Pan.	Average	
											DSC↑	HD95↓
R50 U-Net [8]	30.42	-	85.87	78.19	80.60	63.66	93.74	74.16	87.74	56.90	74.68	36.87
TransUNet [8]	96.07	88.91	85.08	77.02	81.87	63.16	94.08	75.62	87.23	55.86	77.49	31.69
Swin-UNet [7]	27.17	6.16	90.66	79.61	83.28	66.53	94.29	76.60	85.47	56.58	79.13	21.55
MISSFormer [12]	42.46	9.89	91.92	82.00	85.21	68.65	94.41	80.81	86.99	65.67	81.96	18.20
HiFormer-B [11]	25.51	8.05	90.99	79.77	85.23	65.23	94.61	81.08	86.21	59.52	80.39	14.70
Laplacian-Former [3]	27.54	6.68	91.91	80.52	84.23	71.19	94.90	81.14	86.55	64.75	81.90	18.86
WaveFormer [4]	47.01	7.75	91.85	80.06	86.66	69.67	94.43	80.34	85.31	67.08	81.92	18.41
VM-UNet [24]	44.27	6.52	89.51	82.76	86.16	69.41	94.17	81.40	86.40	58.80	81.08	19.21
EW-ViT- w/o Contrastive	33.35	6.02	90.45	80.57	87.68	69.27	94.79	79.62	87.48	69.37	82.40	18.72
EW-ViT	33.35	6.03	93.55	84.57	88.35	69.33	95.17	81.92	88.76	66.43	83.51	16.68

timizer configured with a base learning rate of 0.05, a momentum of 0.9, and a weight decay of 0.0001. The training duration was set to 450 epochs. In contrast, for the ISIC dataset, we used a batch size of 16 and trained the model for 50 epochs with an Adam optimizer at an initial learning rate of 0.0001. The loss function for both the *Synapse* and *ISIC2018* datasets incorporated Boundary Difference Over Union (BDOU) [26] as the segmentation loss ($\mathcal{L}_{Seg}$), combined with a contrastive loss. The overall loss function was defined as $Loss = \lambda_s \mathcal{L}_{Seg} + \lambda_c \mathcal{L}_c$, with $\lambda_s = 1$ and λ_c adjusted dynamically throughout the iterations, as detailed in [11]. For the *ACDC* dataset [6], we trained the models for up to 150 epochs using a batch size of 12 with an Adam optimizer set to a base learning rate of 0.0001. The loss function combined cross-entropy and DICE losses for segmentation ($\mathcal{L}_{seg} = \mathcal{L}_{ce} + \mathcal{L}_{dice}$), in addition to contrastive loss. Data augmentation techniques such as random flipping and rotation were applied across all datasets.

3.3. Results

Synapse Dataset: Table 1 evaluates the performance of our proposed EW-ViT against previous SOTA methods for segmenting abdominal organs in the *Synapse* dataset using DSC and HD95 metrics. EW-ViT significantly outperforms the hybrid CNN-Transformer model HiFormer with a DSC improvement of 3.12%. Additionally, EW-ViT with contrastive supervision outperforms pure transformer-based models like Swin-UNet and FET by 4.38% and 1.59%, respectively. The DSC results highlight EW-ViT’s superiority in segmenting both small (e.g., aorta, kidneys) and large (e.g., spleen, stomach, liver) organs.

Compared to the frequency-based strategies [3, 4], our method achieves better results. Specifically, in [3], the authors build the representation space using the Laplacian pyramid in the frequency domain to adaptively capture multi-resolution representations. Similarly, [4] uti-

lizes wavelets and efficient transformer design to highlight boundary representation. Our method, which focuses on the adaptive and implicit frequency improvement design coupled with contrastive learning, demonstrates significant improvements. This is particularly evident in the case of aorta organs, which require capturing more local patterns to delineate the object of interest. Simultaneously, for large organs, our method better captures long-range dependencies, resulting in improved organ segmentation compared to these frequency-based methods. Additionally, our method achieves the second-best HD95 score. However, HD95 alone does not fully represent performance, especially with noisy annotations.

From a qualitative perspective, we compare our method’s performance to the Waveformer, MissFormer, and TransUNet methods. Moreover, Figure 3 presents visual segmentation results to demonstrate our method’s capabilities. Specifically, our EW-ViT achieves precise boundary segmentation for the left kidney (as shown in the first row), and stomach (second row). These results suggest that our high-frequency filtered features, enriched through implicit prompt-guided interaction, lead to accurate segmentation that surpasses previous architectures.

ISIC2018: Unlike the *Synapse* dataset, where multi-organ segmentation requires modeling anatomical structures, skin lesions do not follow any shape prior information and appear with various textures and deformable structures. Hence, to better delineate skin lesions from dermatology images, methods need to learn inductive biases and local texture patterns. Observing Table 2, we can see that our method achieves superior performance compared to SOTA approaches on DSC and Accuracy metrics. This specifically demonstrates the effectiveness of our method in capturing local and texture patterns from the dataset to detect skin lesions. Our most competitive approach here is the Laplacian-Former, which focuses on a multi-scale frequency approach

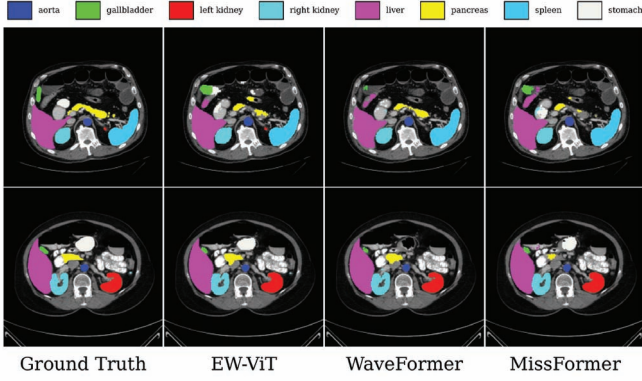


Figure 3. A visual comparison between the EW-ViT and various other techniques using the Synapse dataset.

to better capture local patterns and achieves comparable results to our strategy.

Additionally, the proposed EW-ViT model achieves high Sensitivity ($SE = 0.9297$), which is the best among all methods listed in the table, indicating strong performance in detecting true positives. However, its Specificity ($SP = 0.9571$) is lower compared to some state-of-the-art methods, such as AttU-Net and UCTransNet, which have higher SP values (0.9863 and 0.9825, respectively). This suggests that while our model excels in detecting lesions (high SE), it might have more difficulty in correctly identifying non-lesion areas, potentially leading to over-segmentation. The focus on global contextual information in our model could contribute to this trade-off, as it may affect the refinement of high-frequency components and the accuracy of feature reconstruction in the decoder stages.

Figure 4 further shows EW-ViT’s results from the qualitative point of view comparing to the SOTA approaches. Specifically, our method outperforming hybrid approaches like UCTransNet. These results indicate the excellent generalization ability of our method across different datasets.

Table 2. Performance evaluation of the proposed method against SOTA techniques on the ISIC 2018 benchmark. The best result is highlighted in **blue**, while the second-best result is shown in **red**.

Methods	DSC	SE	SP	ACC
U-Net [23]	0.8545	0.8800	0.9697	0.9404
AttU-Net [21]	0.8566	0.8674	0.9863	0.9376
TransUNet [8]	0.8499	0.8578	0.9653	0.9452
FAT-Net [32]	0.8903	0.9100	0.9699	0.9578
Swin-UNet [7]	0.8946	0.9056	0.9798	0.9645
UCTransNet [27]	0.8838	0.8429	0.9825	0.9527
VM-UNet [24]	0.8971	0.9112	0.9613	0.9491
Laplacian-Former [3]	0.9128	0.9290	0.9626	0.9640
EW-ViT	0.9164	0.9297	0.9571	0.9643

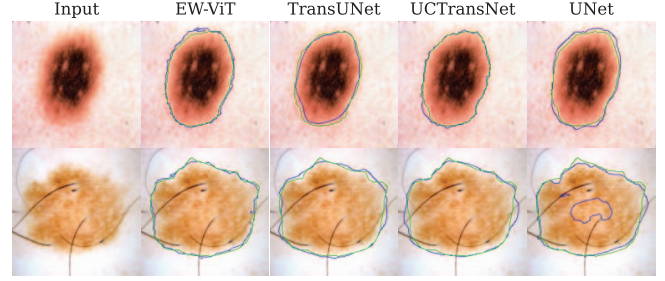


Figure 4. Visual comparisons of various techniques on the ISIC2018 dataset. Grand truth boundaries are represented in **green**, while predicted boundaries are depicted in **blue**.

Table 3. Comparison results of the proposed method assessed on the ACDC dataset are delineated, showcasing DICE scores (%) for individual organs: Right Ventricular (RV), Myocardium (Myo), and Left Ventricular (LV). The optimal outcomes are highlighted in **blue**, with the second-best performances indicated in **red**.

Methods	Avg DICE	RV	Myo	LV
R50+UNet [8]	87.55	87.10	80.63	94.92
R50+AttnUNet [8]	86.75	87.58	79.20	93.47
ViT+CUP [8]	81.45	81.46	70.71	92.18
R50+ViT+CUP [8]	87.57	86.07	81.88	94.75
TransUNet [8]	89.71	88.86	84.53	95.73
Swin-UNet [7]	90.00	88.55	85.62	95.83
MT-UNet [29]	90.43	86.64	89.04	95.62
EW-ViT (Ours)	90.29	88.05	87.56	95.25

ACDC Dataset: Table 3 demonstrates the efficacy of our method comparing to the SOTA approaches on the ACDC dataset. It achieves a notable average DICE score of 90.29%, surpassing TransUNet and SwinUNet by 0.58% and 0.29% respectively, showcasing superior accuracy in cardiac structure delineation. Specifically, our method excels in Myo segmentation with a DICE score of 87.56%, outperforming Swin-UNet and TransUNet by 1.94% and 3.03%, respectively. This highlights its ability to accurately capture intricate cardiac muscle details, crucial for assessing myocardial health. Although our method scores a competitive DICE score of 88.05% in RV segmentation, it remains robust, essential for diagnosing right ventricular conditions. In LV segmentation, our method scores 95.25%, underscoring its effectiveness. However, MT-UNet, a hybrid ViT model that leverages both inter- and intra-affinities feature representation, slightly outperforms our model by 0.14% in overall segmentation. Nevertheless, under degradation conditions, as detailed in Table 4, our EW-ViT designed with the PGHFR module demonstrates superior degradation robustness, surpassing MT-UNet.

Table 4. Comparison results of the different methods against various artifacts during inference time. K indicates Gaussian kernel size. The ACDC dataset was utilized for the entire ablation study.

Method	Gaussian Noise						Gaussian Blur with $K = 115$						Jittering					
	$\sigma = 0.05$		$\sigma = 0.075$		$\sigma = 0.1$		$\sigma = 1$		$\sigma = 1.5$		$\sigma = 1.75$		$\gamma = 0.75$		$\gamma = 1.5$		$\gamma = 1.75$	
	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$	DSC $\uparrow$	$\Delta\downarrow$
Swin-UNet [7]	88.20	1.80	87.40	2.60	84.31	5.69	87.24	2.76	85.49	4.51	83.48	6.52	88.26	1.74	86.12	3.88	82.54	7.46
MT-UNet [29]	88.12	2.31	84.22	6.21	82.09	8.34	86.37	4.06	81.53	8.90	80.13	10.30	84.79	4.06	81.53	8.90	75.53	14.90
EW-ViT	89.52	0.77	87.96	2.33	86.53	3.76	88.60	1.69	86.68	3.61	85.71	4.58	89.31	0.98	87.36	2.93	83.52	6.77

3.4. Robustness to Degradation

To assess the robustness of our approach against degradation, we introduced three types of artifacts to the test set during inference: Gaussian noise at three levels ($\sigma_1 = 0.1$, $\sigma_2 = 0.15$, $\sigma_3 = 0.25$), various blurring configurations (Kernel size = 115, $\sigma_1 = 1$, $\sigma_2 = 1.5$, $\sigma_3 = 1.75$), and color jittering ($\gamma_1 = 0.75$, $\gamma_1 = 1.5$, $\gamma_1 = 1.75$), as summarized in Table 4. Conducting an ablation study on the ACDC Dataset, we evaluated the PGHFR module’s performance under these conditions. Our analysis focused on EW-ViT’s ability to reduce noise in medical images, comparing it with Swin-UNet and MT-UNet methods. Our method consistently outperformed the SOTA approaches, demonstrating less performance degradation as noise levels increased. This underscores our method’s capability to preserve high-frequency information effectively and exhibit robust noise resistance. These findings validate our approach’s efficacy in enhancing the preserving of the high-frequency features.

Furthermore, we extended our evaluation to the *Synapse* dataset with higher noise levels and compared our results with two SOTA approaches Table 5. Our method demonstrated superior robustness to noise effects, exhibiting a smaller performance drop compared to the other methods. This highlights the robustness and reliability of our approach under challenging conditions. We also visually showcase the effect of Gaussian noise on the input image and visualize the attention map of our method to demonstrate that it effectively focuses on the object of interest even at high noise levels, as illustrated in Figure 5. This visual analysis further reinforces our method’s capability to maintain accurate segmentation despite noise interference. Finally, In Figure 6, we also illustrate the impact of noise at various sigma values on each model to identify the point at which each model degrades.

4. Conclusion

Our method addresses ViT limitations in medical image segmentation by integrating wavelet-based frequency techniques and a PGHFR module, enhancing representations, reducing complexity, and improving noise robustness. Contrastive learning further ensures feature consistency, making the approach highly resilient to noise.

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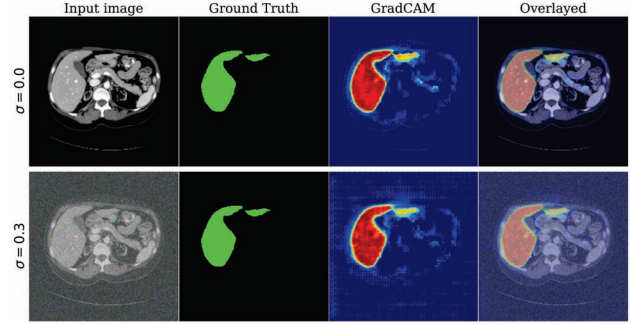


Figure 5. Visualization of attention maps generated by our proposed method under noisy conditions.

Table 5. Noise robustness comparison between EW-ViT and the previous methods on the *Synapse* dataset.

Gaussian Noise	Methods	DSC $\uparrow$	$\Delta\downarrow$
$\sigma = 0.1$	TransUNet [8]	75.75	1.74
	WaveFormer [4]	80.66	1.26
	EW-ViT	83.40	0.11
$\sigma = 0.15$	TransUNet [7]	72.68	4.81
	WaveFormer [4]	78.57	3.35
	EW-ViT	82.07	1.44
$\sigma = 0.25$	TransUNet [7]	67.87	9.62
	WaveFormer [4]	74.41	7.51
	EW-ViT	79.70	3.81

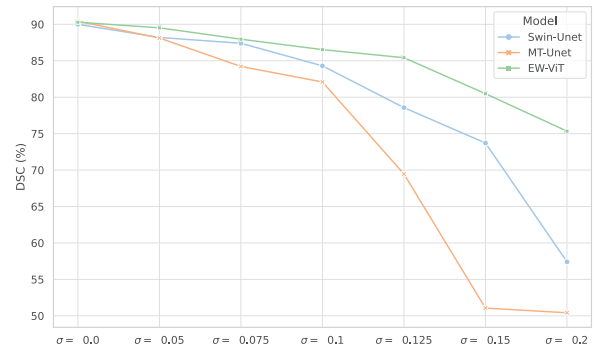


Figure 6. Noise impact analysis using the ACDC dataset.

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References

- [1] Reza Azad, Maryam Asadi-Aghbolaghi, Mahmood Fathy, and Sergio Escalera. Bi-directional convlstm u-net with densley connected convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision workshops*, pages 0–0, 2019. 5
- [2] Reza Azad, Moein Heidari, Yuli Wu, and Dorit Merhof. Contextual attention network: Transformer meets u-net. In *Machine Learning in Medical Imaging: 13th International Workshop, MLMI 2022, Held in Conjunction with MICCAI 2022, Singapore, September 18, 2022, Proceedings*, pages 377–386. Springer, 2022. 1
- [3] Reza Azad, Amirhossein Kazerooni, Babak Azad, Ehsan Khodapanah Aghdam, Yury Velichko, Ulas Bagci, and Dorit Merhof. Laplacian-former: Overcoming the limitations of vision transformers in local texture detection. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 736–746. Springer, 2023. 6, 7
- [4] Reza Azad, Amirhossein Kazerooni, Alaa Sulaiman, Afshin Bozorgpour, Ehsan Khodapanah Aghdam, Abin Jose, and Dorit Merhof. Unlocking fine-grained details with wavelet-based high-frequency enhancement in transformers, 2023. 2, 6, 8
- [5] Jiawang Bai, Li Yuan, Shu-Tao Xia, Shuicheng Yan, Zhifeng Li, and Wei Liu. Improving vision transformers by revisiting high-frequency components. In *European Conference on Computer Vision*, pages 1–18. Springer, 2022. 2
- [6] Olivier Bernard, Alain Lalande, Clement Zotti, Frederick Cervenansky, Xin Yang, Pheng-Ann Heng, Irem Cetin, Karim Lekadir, Oscar Camara, Miguel Angel Gonzalez Ballester, et al. Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: is the problem solved? *IEEE transactions on medical imaging*, 37(11):2514–2525, 2018. 6
- [7] Hu Cao, Yueyue Wang, Joy Chen, Dongsheng Jiang, Xiaopeng Zhang, Qi Tian, and Manning Wang. Swin-unet: Unet-like pure transformer for medical image segmentation. In *Proceedings of the European Conference on Computer Vision Workshops(ECCVW)*, 2022. 6, 7, 8
- [8] Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L Yuille, and Yuyin Zhou. Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*, 2021. 1, 5, 6, 7, 8
- [9] Noel Codella, Veronica Rotemberg, Philipp Tschandl, M Emre Celebi, Stephen Dusza, David Gutman, Brian Helba, Aadi Kalloo, Konstantinos Liopyris, Michael Marchetti, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1902.03368*, 2019. 5
- [10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 1
- [11] Moein Heidari, Amirhossein Kazerooni, Milad Soltany, Reza Azad, Ehsan Khodapanah Aghdam, Julien Cohen-Adad, and Dorit Merhof. Hiformer: Hierarchical multi-scale representations using transformers for medical image segmentation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 6202–6212, 2023. 1, 6
- [12] Xiaohong Huang, Zhifang Deng, Dandan Li, Xueguang Yuan, and Ying Fu. Missformer: An effective transformer for 2d medical image segmentation. *IEEE Transactions on Medical Imaging*, pages 1–1, 2022. 2, 6
- [13] Yuhao Huang, Xin Yang, Han Zhou, Yan Cao, Haoran Dou, Fajin Dong, and Dong Ni. Robust box prompt based sam for medical image segmentation. *arXiv preprint arXiv:2407.21284*, 2024. 4
- [14] Salman Khan, Muzammal Naseer, Munawar Hayat, Syed Waqas Zamir, Fahad Shahbaz Khan, and Mubarak Shah. Transformers in vision: A survey. *ACM Computing Surveys*, 54(10s):1–41, Jan. 2022. 1
- [15] Bennett Landman, Zhoubing Xu, J Igelsias, Martin Styner, T Langerak, and Arno Klein. Miccai multi-atlas labeling beyond the cranial vault—workshop and challenge. In *Proc. MICCAI Multi-Atlas Labeling Beyond Cranial Vault—Workshop Challenge*, volume 5, page 12, 2015. 5
- [16] Xiangru Li, Yifei Zhang, and Liang Zhao. Multi-prompt fine-tuning of foundation models for enhanced medical image segmentation. *arXiv preprint arXiv:2310.02381*, 2023. 4
- [17] Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation, 2021. 4
- [18] Lin Liu, Jianzhuang Liu, Shanxin Yuan, Gregory Slabaugh, Ales Leonardis, Wengang Zhou, and Qi Tian. Wavelet-based dual-branch network for image demoreing, 2020. 3
- [19] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 1
- [20] Yao Lu, Bo Jiang, Guangming Lu, and Bob Zhang. Implicit prompt learning for image denoising. 33(1):4678–4686, 2024. 4
- [21] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y Hammerla, Bernhard Kainz, et al. Attention u-net: Learning where to look for the pancreas. *arXiv preprint arXiv:1804.03999*, 2018. 7
- [22] Vaishnav Potlapalli, Syed Waqas Zamir, Salman Khan, and Fahad Shahbaz Khan. Promptir: Prompting for all-in-one blind image restoration, 2023. 4
- [23] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18*, pages 234–241. Springer, 2015. 1, 7

- [24] Jiacheng Ruan and Suncheng Xiang. Vm-unet: Vision mamba unet for medical image segmentation. *arXiv preprint arXiv:2402.02491*, 2024. 6, 7
- [25] Kihyuk Sohn, Yuan Hao, José Lezama, Luisa Polania, Huiwen Chang, Han Zhang, Irfan Essa, and Lu Jiang. Visual prompt tuning for generative transfer learning, 2022. 4
- [26] Fan Sun, Zhiming Luo, and Shaozi Li. Boundary difference over union loss for medical image segmentation, 2023. 6
- [27] Haonan Wang, Peng Cao, Jiaqi Wang, and Osmar R Zaiane. Uctransnet: rethinking the skip connections in u-net from a channel-wise perspective with transformer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 36, pages 2441–2449, 2022. 7
- [28] Haixin Wang, Jianlong Chang, Yihang Zhai, Xiao Luo, Jinan Sun, Zhouchen Lin, and Qi Tian. Lion: Implicit vision prompt tuning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 5372–5380, 2024. 4
- [29] Hongyi Wang, Shiao Xie, Lanfen Lin, Yutaro Iwamoto, Xian-Hua Han, Yen-Wei Chen, and Ruofeng Tong. Mixed transformer u-net for medical image segmentation. In *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 2390–2394. IEEE, 2022. 7, 8
- [30] Peihao Wang, Wenqing Zheng, Tianlong Chen, and Zhangyang Wang. Anti-oversmoothing in deep vision transformers via the fourier domain analysis: From theory to practice. In *International Conference on Learning Representations*, 2022. 2
- [31] Wenxuan Wang, Chen Chen, Meng Ding, Hong Yu, Sen Zha, and Jiangyun Li. Transbts: Multimodal brain tumor segmentation using transformer. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part I 24*, pages 109–119. Springer, 2021. 1
- [32] Huisi Wu, Shihuai Chen, Guilian Chen, Wei Wang, Baiying Lei, and Zhenkun Wen. Fat-net: Feature adaptive transformers for automated skin lesion segmentation. *Medical Image Analysis*, 76:102327, 2022. 7
- [33] Zhuofan Xia, Xuran Pan, Shiji Song, Li Erran Li, and Gao Huang. Vision transformer with deformable attention, 2022. 1
- [34] Yutong Xie, Jianpeng Zhang, Chunhua Shen, and Yong Xia. Cotr: Efficiently bridging cnn and transformer for 3d medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part III 24*, pages 171–180. Springer, 2021. 1
- [35] Ting Yao, Yingwei Pan, Yehao Li, Chong-Wah Ngo, and Tao Mei. Wave-vit: Unifying wavelet and transformers for visual representation learning. In *Computer Vision–ECCV 2022: 17th European Conference, Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part XXV*, pages 328–345. Springer, 2022. 2
- [36] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Conditional prompt learning for vision-language models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16816–16825, 2022. 4
- [37] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9):2337–2348, 2022. 4
- [38] Yanfeng Zhou, Jiaxing Huang, Chenlong Wang, Le Song, and Ge Yang. Xnet: Wavelet-based low and high frequency fusion networks for fully- and semi-supervised semantic segmentation of biomedical images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 21085–21096, October 2023. 2